

The influence of heavy metal stress on the evolutionary transition of teosinte to maize

Reviewed Preprint

v1 • June 4, 2025

Not revised

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eLife Assessment

This study presents a **valuable** investigation into how heavy metal stress may have influenced the domestication of maize from its wild ancestor, teosinte *parviglumis*, focusing on specific ATPase genes with proposed roles in heavy metal homeostasis. The evidence supporting the main claims is **incomplete**, with suggestive but not definitive data linking gene function to domestication traits, and limited environmental context for the hypothesized selection pressures. While the work introduces an interesting model connecting environmental stress responses to evolutionary transitions and highlights underexplored aspects of teosinte plasticity, the conclusions would benefit from more comprehensive analyses such as transcriptomics, a broader survey of loci, and stronger paleoenvironmental validation. The study will be of interest to researchers in plant evolution and domestication, but currently lacks the analytical depth to fully support its central hypothesis.

<https://doi.org/10.7554/eLife.105858.1.sa3>

Abstract

Maize originated from teosinte *parviglumis* following a subspeciation event occurred in volcanic regions of Mesoamerica. The elucidation of the phenotypic changes that gave rise to maize have focused on the direct consequences of domestication, with no insights on how environmental factors could have influenced specific gene function and human selection. We explored the impact of heavy metal (HM) stress by exposing both subspecies to sublethal concentrations of copper and cadmium. We also assessed the genetic diversity of loci encompassing three HM response genes affected by domestication: *ZmHMA1*, *ZmHMA7* – encoding for heavy metal ATPases of the P1_b family-, and *ZmSKUs5*, encoding a multicopper oxidase. *ZmHMA1* and *ZmSKUs5* map in the short arm of chromosome five, in a genomic region containing multiple linked QTLs with pleiotropic effects on domestication. A genomic analysis of the full chromosome shows that their loci show strong positive selection as compared to previously identified domestication genes. Exposure of teosinte *parviglumis* to HM stress results in a plant architecture reminiscent of extant maize, and upregulation of *Teosinte branched1 (Tb1)* in the meristem. *ZmHMA1* and *ZmHMA7* are expressed throughout

development and respond to HM stress in both subspecies. *ZmHMA1* is mainly involved in restricting plant height and optimizing the number of female inflorescences and seminal roots. Our results suggest that HM stress acted on specific ATPases involved in homeostasis, giving rise to phenotypic variants that were identified and selected by humans during domestication.

Introduction

Driven by either conscious or unconscious human selection, domestication can be considered a gradual evolutionary process resulting in the adaptation to agroecological environments and anthropogenic preferences (Ross-Ibarra et al. 2007 [↗](#)). In the case of plants, the obvious consequences of domestication are represented by visually recognizable traits that include changes in photoperiod response, reduced capacity of seed dispersion, reduced lateral branching, larger fruits or grains, reduced duration of seed dormancy, and increase in the size of female inflorescences (Pickersgill 2007 [↗](#); Larson et al. 2014 [↗](#)). Notably, research efforts to understand the phenotypic changes that gave rise to maize have mainly focused on the direct consequences of domestication in aerial organs and sometimes roots (Chen et al. 2022 [↗](#); Lopez-Valdivia et al. 2022 [↗](#); Ren et al. 2022 [↗](#)), with no emphasis on the role of adaptive environmental responses that, by acting on specific gene function, could have given rise to phenotypic traits that were identified and selected by humans.

It is generally accepted that domestication of maize (*Zea mays* ssp. *mays*) from teosinte *parviglumis* (*Zea mays* ssp. *parviglumis*) initiated in Mesoamerica around 9,000 years before present (Matsuoka et al. 2002 [↗](#)). The initial evolutionary transition that resulted in the event of subspeciation most likely occurred within the Balsas river drainage, in regions geographically converging with the Trans-Mexican Volcanic Belt, at the intersection of the States of Mexico, Guerrero, and Michoacan (Matsuoka et al. 2002 [↗](#)). Although the nature of the initial phenotypic changes that allowed the emergence of primitive maize remains unclear, the most obvious architectural and morphological differences between both subspecies affect their vegetative and reproductive organs. Teosinte *parviglumis* generates numerous lateral branches ending in male inflorescences, while maize often has a few short or no lateral branches, not ending in male reproductive organs (Wills et al. 2013 [↗](#)). Additionally, the seeds of teosinte *parviglumis* are covered by a hard glume that protects the kernel externally, while in maize glumes are reduced, soft and do not cover the kernel (Doebley and Stec 1993 [↗](#)). Large scale efforts involving physical mapping and comparative genomic analysis have resulted in the identification of numerous regions showing differential nucleotide variability between teosinte *parviglumis* and the B73 reference maize genome. In some cases, mutations in genes located within those genomic regions result in phenotypes that reveal their involvement in domestication traits. One of the major genes affected by maize domestication is *Teosinte branched1* (*Tb1*), a transcription factor of the TCP family that controls lateral branching, sexual determination, and glume formation (Doebley et al. 1997 [↗](#); Martín-Trillo and Cubas 2010 [↗](#)). *Tb1* acts as a repressor of lateral branching (Studer et al. 2017 [↗](#)), giving rise to maize plants usually composed of a single stem. Other genes involved in domestication include *Teosinte glume architecture1* (*Tga1*), *ramosa2* (*ra2*) and *Zea floricaula leafy2* (*zfl2*), among others (Bomblies and Doebley 2006 [↗](#); Sigmon and Vollbrecht 2010 [↗](#); Wang et al. 2015 [↗](#)).

The importance of phenotypic plasticity is considered as likely relevant in the domestication history of crops (Ross-Ibarra et al. 2007 [↗](#); Lorant et al. 2020 [↗](#); Mueller et al., 2023); however, the role of environmental factors influencing their origins remains elusive and poorly investigated. A previous study showed that teosinte *parviglumis* grown under low temperatures and CO₂ concentrations - reminiscent of those prevailing in the late Pleistocene and early Holocene - exhibit some phenotypes related to the domestication syndrome, including short branches, a single main stalk, and partially open seed fruitcases (Piperno et al. 2015 [↗](#); Piperno et al. 2019); however, the genetic basis of these changes have not been investigated. Although these results

suggest that phenotypic plasticity could have influenced the domestication process, to this date there are no hints on how the adaptation to local environmental conditions could have affected specific gene function and guided human selection.

Recent genomic evidence indicates that admixture of maize with native teosinte *mexicana* populations (*Zea mays* ssp. *mexicana*) contributed to maize adaptation across a wide range of altitudes and latitudes in the American continent (Yang et al. 2023 [\[1\]](#)); however, the earliest maize macro-specimens found to date lack teosinte *mexicana* admixture and exhibit morphological attributes of fully domesticated cobs, indicating that *mexicana* is not a direct ancestor of maize and that its introgression occurred after initial divergence from teosinte *parviglumis* (Vallebuena-Estrada et al. 2023 [\[2\]](#); Acosta-Bayona et al. 2024 [\[3\]](#)). Consequently, the nature of the genetic factors that guided the emergence of maize remain largely unknown, and the initial phenotypic changes that caused the original divergence of these two subspecies have not been identified. It is also unknown if domestication initiated by direct action of human selection on teosinte natural standing variation, or if the response to specific environmental factors generated phenotypic changes that were identified and selected by humans to act on native teosinte *parviglumis* populations.

Maize can adapt to abiotic factors such as heavy metal (HM) stress through numerous mechanisms that prevail during development (Abdelgawad et al. 2020 [\[4\]](#); Gallo-Franco et al. 2020 [\[5\]](#); Malenica et al. 2021 [\[6\]](#); Gao et al. 2022 [\[7\]](#)). These include metal immobilization by mycorrhizal association, metal sequestration, or complexation by exuding organic compounds from roots, metal intracellular sequestration and compartmentalization in vacuoles. HMs include essential elements that are necessary for maize growth through enzymatic function, such as iron (Fe), manganese (Mn), zinc (Zn), magnesium (Mg), molybdenum (Mo), and copper (Cu); and non-essential elements like Cadmium (Cd), chromium (Cr), lead (Pb), aluminum (Al), and selenium (Se). In general, HM stress leads to visibly reduced plant growth due to the reduced cell elongation and cell wall elasticity. At the molecular level, metal toxicity can disturb the cellular redox balance and lead directly or indirectly to oxidative stress and the formation of reactive oxygen species (ROS), shifting the redox balance to the pro-oxidative side until the antioxidative defense system allows a new balanced redox status (Smeets et al. 2009 [\[8\]](#); Hall 2002 [\[9\]](#)).

The comparison of the *Palomero toluqueño* and B73 genomes resulted in the identification of a large collection of identical sequence regions (IdSRs) with low nucleotide variability and transcriptional evidence of expression (Vielle-Calzada et al. 2009 [\[10\]](#)). Putative domestication loci included *ZmHMA1* and *ZmHMA7*, two genes encoding members of the P_{1b}-ATPases protein superfamily, also known as Heavy Metal ATPases (P-type ATPases) characterized by the generation of a phosphorylated intermediate during the transport reaction cycle. Members of this superfamily contain 8 to 12 transmembrane domains as well as conserved ATP binding, phosphorylation and dephosphorylation sites (Solioz and Vulpe 1996 [\[11\]](#); Axelsen and Palmgren 2001 [\[12\]](#); Voskoboinik et al. 2003 [\[13\]](#); Williams and Mills 2005 [\[14\]](#)). The superfamily can be divided in subfamilies that include H⁺ATPases (type 3_A) in plants and fungi, Na⁺/K ATPases (type 2_{C/D}) in animals, Ca²⁺ATPases (type 2_{A/B}), and heavy metal (HM) transporting ATPases (type 1_B) capable of transporting a large variety of HM cations across cell membranes. *ZmHMA1* and *ZmHMA7* are type 1_B ATPases that include a conserved Cys-Pro-Cys/His/Ser motif within a transmembrane domain necessary for the translocation of HMs cations through the membrane channel. A third *Palomero toluqueño*/B73 IdSR included *Skewed Root Growth Similar5* (*ZmSKU5*), a gene encoding a Skewed5 (SKU5)-similar (SKS) protein classified as a multi-copper oxidase containing cupredoxin domains (Sedbrook et al. 2002 [\[15\]](#)). All three genes map to chromosome five (chr.5), with *ZmHMA1* and *ZmSKU5* within a genomic region that fractionates into multiple QTLs under selection during the *parviglumis* to maize transition (Doebley and Stec 1993 [\[16\]](#); Lemmon and Doebley, 2014). Contrary to major domestication loci showing a single highly pleiotropic gene responsible for important domestication traits, in this chr.5 genomic region the smallest 1.5-LOD support interval contained more than 50 genes, suggesting that phenotypic effects are due to multiple linked QTLs

(Lemmon and Doebley, 2014). While the functional characterization of all three maize genes remains elusive, the Arabidopsis and barley *ZmHMA1* homologs act during heavy metal transport in the chloroplast (Boutigny et al. 2014 [↗](#)), demonstrating their functional involvement in homeostasis.

To investigate a possible involvement of heavy metal response in the origin and domestication of maize, we compared the phenotypic changes that teosinte *parviglumis* and maize undergo when grown in the presence of sublethal concentrations of Cu and Cd. We show that teosinte *parviglumis* responds to HM stress by acquiring an aerial architecture reminiscent of extant maize, and that *Tb1* expression is upregulated in teosinte *parviglumis* meristems when grown under HM stress. A large-scale genomic analysis of nucleotide variability across maize chr.5 indicates that loci encompassing *ZmHMA1*, *ZmHMA7*, and *ZmSKUs5* show reduced nucleotide variability in maize as compared to teosinte *parviglumis*. Both *ZmHMA1* and *ZmHMA7* are active throughout maize and teosinte *parviglumis* development, and their expression responds to HM stress. In addition, mutations in *ZmHMA1* show phenotypic defects related to the maize domestication syndrome, some of which are only manifested under exposure to HMs. Overall, our results indicate that the evolutionary transition that gave rise to extant maize was influenced by heavy metal stress through the activity of P1b-ATPases.

Results

Teosinte *parviglumis* responds to heavy metal stress by acquiring an architecture reminiscent of extant maize

To compare HM response in teosinte *parviglumis* and maize, we exposed wild-type W22 maize inbred and CIMMYTMA 29791 teosinte *parviglumis* individuals to sublethal concentrations of Cu (400 mg/kg of CuSO_4) and Cd (16mg/kg of CdCl_2). CuSO_4 and CdCl_2 were homogenized in soil before transplant and grown under controlled greenhouse conditions. HM concentrations were based on preliminary dose responses that allowed maize plants to reach reproductive stages and complete their life cycle (Abdelgawad et al. 2020 [↗](#)). As previously reported (Atta et al. 2023 [↗](#)), maize individuals grown in the presence of HMs showed a significant reduction in aerial traits such as stem diameter, size of leaves, size and number of roots, and size of female inflorescences (**Figure 1** [↗](#); **Tables 2** [↗](#) and **S2** [↗](#)). On average, stem diameter showed a reduction of about 17% as compared to plants grown in absence of HM stress (**Figure 1f** [↗](#) to **1j** [↗](#)). Similarly, the length and width of developing leaves showed average reductions of 12%, and mature leaves of 6%. Reproductive traits such as the number of female inflorescences (NFI) and their average weight (WFI) showed a significant reduction (NFI = 1.20 ± 0.42 in absence of HMs and NFI = 0.70 ± 0.67 in presence of HMs; WFI = 40.06 ± 14.39 in absence of HMs and WFI = 37.75 ± 18.83 in presence of HMs; **Table 2** [↗](#)). As expected, the most significant effects occurred in the root system, with a significant increase in the number of brace roots under HMs (NBR = 12.90 ± 4.33 in absence of HMs; NBR = 16.30 ± 4.99 in presence of HMs; **Table 2** [↗](#)), and a significant decrease in their length (LBR = 16.15 ± 9.75 cm in absence of HMs; LBR = 4.00 ± 2.04 cm in presence of HMs; **Table 2** [↗](#)). The number and length of crown roots were reduced on average by 33% and 18%, respectively (NCR = 11.30 ± 2.11 in absence of HMs; NCR = 7.60 ± 0.97 in presence of HMs; LCR = 58.90 ± 21.87 cm in absence of HMs; LCR = 48.85 ± 11.80 cm in presence of HMs; **Table 2** [↗](#)). Nodal roots also showed a significant reduction in their number and length (NNR = 10.50 ± 2.27 in absence of HMs; NNR = 6.70 ± 2.79 in presence of HMs; LNR = 87.20 ± 16.80 cm in absence of HMs; LNR = 55.30 ± 21.10 cm in presence of HMs; **Tables 2** [↗](#) and **S2** [↗](#)). By contrast, the number (NSR) and length (LSR) of seminal roots was significantly increased in the presence of HMs (NSR = 4.10 ± 1.45 in absence of HMs; NSR = 6.60 ± 1.35 in presence of HMs; LSR = 27.70 ± 13.38 cm in absence of HMs; LSR = 43.50 ± 14.01 cm in presence of HMs) when measured 50 days after transplant (**Table 2** [↗](#)).

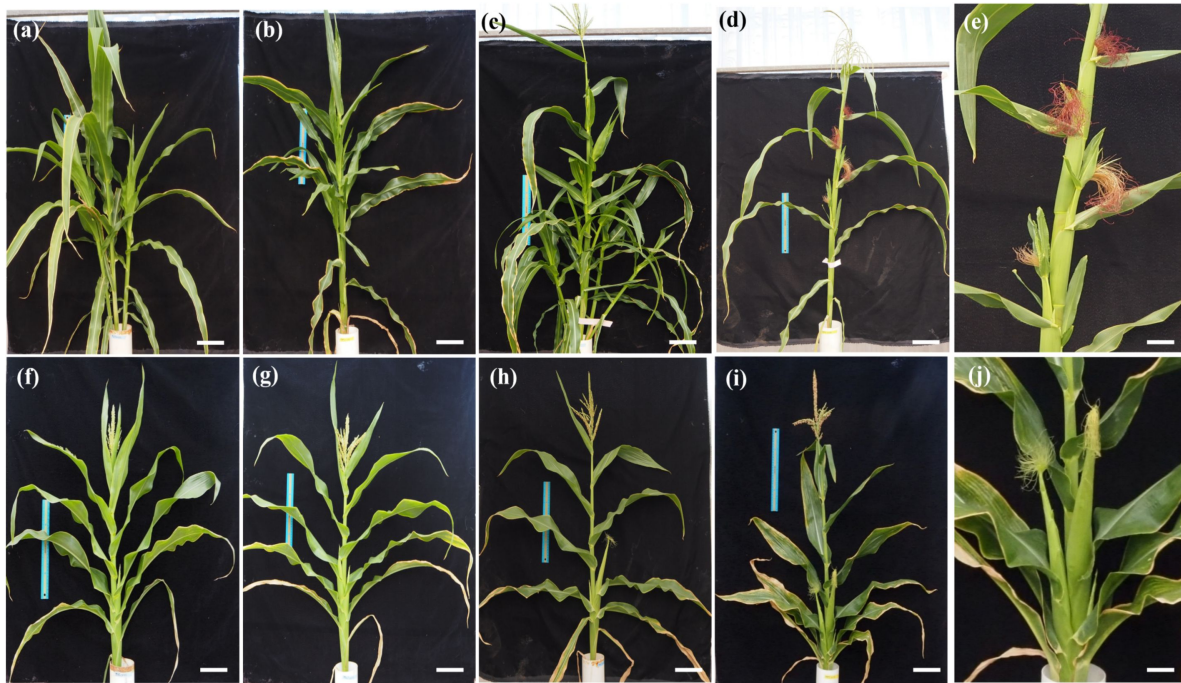


Figure 1.

Phenotypic response of teosinte *parviglumis* and wild-type W22 maize grown under heavy metal stress.

(a) to (e), teosinte *parviglumis*; (f) to (j), W22 maize. (a) Teosinte *parviglumis* in absence of heavy metal stress at early flowering stage. (b) Teosinte *parviglumis* grown under heavy metal stress at early flowering stage. (c) Teosinte *parviglumis* in absence of heavy metal stress at late flowering stage. (d) Teosinte *parviglumis* grown under heavy metal stress at late flowering stage. (e) Teosinte *parviglumis* showing proliferation of female inflorescences arising from independent axillary meristems under heavy metal stress (detail of d). (f) W22 maize in absence of heavy metal stress at early flowering stage. (g) W22 maize grown under heavy metal stress at early flowering stage. (h) W22 maize in absence of heavy metal stress at late flowering stage. (i) W22 maize grown under heavy metal stress at late flowering stage. (j) W22 maize female inflorescences at late flowering stage (detail of i). Scale bars: 10 cm.

Interestingly, teosinte *parviglumis* exhibited a maize-like phenotype when grown in the presence of HMs, with complete absence of tillering in most plants (**Figure 1** [↗](#); **Tables 1** [↗](#), **S1** [↗](#) and **S3** [↗](#)). From a total of 30 plants tested, 28 showed total absence of tillering, and two initiated secondary stems during vegetative stages but arrested their growth early during development, resulting in the presence of a single dominant stem at reproductive stages (**Figure 1a** [↗](#) to **1d** [↗](#)). Additionally, all 30 plants tested showed complete absence or residual lateral branching, with unusual proliferation of female inflorescences arising from independent axillary meristems in primary stems (**Figure 1e** [↗](#)). This significant increase in the number of female inflorescences clustered within a single stem ($NFI = 1.80 \pm 0.63$ in absence of HMs; $NFI = 3.90 \pm 1.66$ in presence of HMs; **Figure 1e** [↗](#) and **Table 1** [↗](#)) was associated with shorter flowering time as compared with normal growing conditions in absence of HMs (**Figure 1d** [↗](#)). In addition, teosinte *parviglumis* grown in the presence of HMs showed significantly reduced leaf length (LIL = 71.30 ± 11.32 cm in absence of HMs; 56.25 ± 7.80 cm in presence of HMs; **Table 1** [↗](#)). The absence of lateral branches resulted in lower values of aerial dry weight (DWAP = 133.49 ± 28.28 gr in the absence of HMs; 82.72 ± 11.67 gr in presence of HMs; **Table 1** [↗](#)). Teosinte *parviglumis* grown under HMs showed a significant reduction in the length of brace roots (LBR = 20.00 ± 11.49 cm in absence of HMs; 2.88 ± 0.66 cm in presence of HMs; **Table 1** [↗](#)), but no significant changes in their number. The number and length of crown and nodal roots was not affected by HM stress, nor did the presence of HMs modify the absence of seminal roots intrinsic to teosinte *parviglumis* accessions (Perkins and Lynch 2021 [↗](#)). Taken together, these results suggest that the response of maize and teosinte *parviglumis* to HM stress can give rise to phenotypes previously shown to distinguish both subspecies at the light of the domestication syndrome.

Genomic regions encompassing *ZmHMA1*, *ZmHMA7* and *ZmSKUs5* show reduced nucleotide variability in extant maize as compared to teosinte *parviglumis*

ZmHMA1 and *ZmSKU5* mapped 52.1 and 41.8 cM, respectively, in a genetic map of chr.5 calculated to be 86.64 cM, with an average megabase pair to centimorgan ratio of 1.873 Mbp/cM (Lemmon and Doebley, 2014; **Figure S1** [↗](#)). The corresponding mapping population used isogenic recombinant inbred lines (NIRILs) with informative cross-over events concentrated in the region of interest and replicated phenotypic assessments in block experiments. Multiple QTLs were detected in a region spanning across 82 cM, none with single large effect genes such as those found in other chromosomes (Wang et al., 2005; Studer et al. 2011; Wills et al., 2013 [↗](#)), suggesting that several associated domestication traits are due to multiple linked QTLs of small effect (Lemmon and Doebley, 2014). The location of *ZmSKU5* overlaps with a single wide 1.5-LOD support interval that spreads across 50.6 cM (number of kernels per rank; **Figure S1** [↗](#)). By contrast, *ZmHMA1* overlaps with a contiguous region containing no less than five QTL 1.5- LOD support intervals (ear diameter; number of kernels per rank; percentage of staminate spikelets; tassel branch number; and tillering index; **Figure S1** [↗](#)).

To determine if the regulatory and coding regions of *ZmHMA1*, *ZmHMA7*, and *ZmSKUs5* were affected by human selection, we calculated their level of nucleotide variability based on single nucleotide polymorphisms (SNPs) for all extant maize and teosinte *parviglumis* accessions included in the diversity panel HapMap3, using the corresponding sequences of *Tripsacum dactyloides* as an outgroup. The levels of the nucleotide variability index - $\pi(m)$ and $\pi(t)$ for maize and teosinte *parviglumis*, respectively - were determined in bins of 100 nt using VCFtools, including alignments encompassing a genomic region that included 15 kb upstream of the START codon, the corresponding coding region for each gene, and 15 kb downstream of the STOP codon (**Figure 2** [↗](#) and **Table S4** [↗](#)). In the case of *ZmHMA1*, the upstream region values were $\pi(m)=0.00071$ and $\pi(t)=0.00870$, and for the downstream region, $\pi(m)=0.00097$ and $\pi(t)=0.00248$. By contrast, the *ZmHMA1* coding sequence showed low values in both subspecies: $\pi(m)=0.0050$ and $\pi(t)=0.00106$. These results indicate that human selection in the *ZmHMA1* locus preferentially

		Description	Cu/Cd(-)		Cu/Cd(+)		P Value
			Mean	SD	Mean	SD	
Aerial traits	LIL	Length of Immature Leaf (cm)	71.30	11.92	56.25	7.80	0.0570
	SB	Stem Branches	5.30	1.83	0.60	0.84	0.0002***
	DWAP	Dry Weight of Aerial Part (g)	133.49	28.28	82.72	11.67	0.7430
	NFI	Number of Female Inflorescences	1.80	0.63	3.90	1.66	0.0670
	LFI	Length of Female Inflorescences (cm)	15.40	1.35	12.50	2.12	0.0650
Root traits	NBR	Number of Brace Roots	10.60	6.13	11.60	4.55	0.1210
	LBR	Length of Brace Roots (cm)	20.00	11.49	2.88	0.66	0.0045**
	NCR	Number of Crown Roots	12.80	4.34	9.60	1.43	0.0002***
	LCR	Length of Crown Roots (cm)	60.70	16.95	62.80	14.10	0.2219
	NNR	Number of Nodal Roots	16.20	4.69	14.00	3.20	0.0039**
	LNR	Length of Nodal Roots (cm)	83.00	15.76	104.00	8.46	0.0016***

** P<0.01; *** P<0.001.

Table 1.

Estimation of phenotypic traits in teosinte *parviglumis* grown in absence and presence of heavy metal stress.

affected its upstream and downstream regulatory regions (**Figure 2a** and **Table S4**). Interestingly, although maize π values are at least one order of magnitude lower than π values in teosinte, $\pi(t)$ values are also one order of magnitude lower than π values of *Tripsacum sp.*, suggesting that a genetic bottleneck also affected teosinte *parviglumis* in these loci. *ZmHMA7* showed contrasting values of $\pi(m)$ and $\pi(t)$ across the entire coding sequence and its regulatory regions (**Figure 2b**). For the upstream region, values were $\pi(m) = 0.0009$ and $\pi(t) = 0.00411$; for the *ZmHMA7* coding region, $\pi(m) = 0.00015$ and $\pi(t) = 0.00148$; and for the downstream region, $\pi(m) = 0.00066$ and $\pi(t) = 0.00454$. *ZmSKUs5* also showed contrasting nucleotide variability in the entire coding and flanking regions (**Figure 2c** and **Table S4**). Values were $\pi(m) = 0.0011$ and $\pi(t) = 0.00735$ for the upstream region, $\pi(m) = 0.00044$ and $\pi(t) = 0.00754$ for the coding sequence, and $\pi(m) = 0.00203$ and $\pi(t) = 0.00626$ for the downstream region (**Table S4**). For all three genes, similar π values in the coding region of maize landraces and improved lines indicates that the loss of nucleotide variability occurred during the teosinte *parviglumis* to maize transition, and not during subsequent contemporary breeding (**Table S4**). By contrast, the *ZmGLB1* neutral control locus shows equivalent levels of nucleotide variability in maize and teosinte *parviglumis* (Eyre-Walker et al. 1998).

We also compared the level of genetic variability in *ZmHMA1*, *ZmSKUs5*, and *ZmHMA7* (and their neighboring regions) to all chr.5 genes categorized as likely affected by domestication (Hufford et al., 2012), and to the full collection of neutral genes distributed across the whole chromosome. The analysis was conducted using 100 bp bins and sliding windows of 50 bp; for a total of 385,829 bins in the case of chr.5 domestication genes, and 2,373,010 bins for neutral genes. As shown in **Figure S2**, the levels of nucleotide variability in the three candidate genes and their six neighboring loci are significantly reduced in maize landraces as compared to teosinte *parviglumis* inbred lines. When compared to the full neutral gene set of chr.5, this reduction is more severe than the reduction affecting 304 previously identified domestication genes, confirming the exceptional selective pressure that was imposed in those three loci during the evolutionary transition from teosinte *parviglumis* to extant maize.

ZmHMA1 and ZmHMA7 are active throughout development and their expression is influenced by the presence of heavy metals

The gene structure of *ZmHMA1*, *ZmHMA7*, and *ZmSKUs5* is shown in **Figure 3**. The coding sequence of *ZmHMA1* and *ZmHMA7* is approximately 21.6 Kb and 4.5 Kb, respectively, while *ZmSKUs5* is approximately 4.2 Kb in length. Within the short arm of chromosome 5, *ZmHMA1* and *ZmSKUs5* are separated by 9.16 Mb, with *ZmHMA1* approximately 15 Mb away from the maximum likelihood prediction of a QTL involved in maize domestication (Doebley and Stec 1993; Villeda-Calzada et al. 2009). *ZmHMA7* maps to the centromeric region of chromosome 5. A detailed PlantCARE-based analysis identified several conserved binding sites and regulatory elements in the promoter region of these three genes (Lescot et al. 2002). *ZmHMA1* and *ZmHMA7* share binding sites to basic Helix-Loop-Helix (bHLH) transcription factors, as well as abscisic acid response (ABRE), ethylene responsive (ERF), and G-box (CACGTG) elements recognized by leucine zipper family (bZIP) and bHLH transcription factors (**Figure 3a**). The regulatory region of both *ZmHMA1* and *ZmSKUs5* contain a binding site for M1KC-type MADS box transcription factors (**Figure 3a**), whereas *ZmHMA7* and *ZmSKUs5* share a jasmonic acid-response TGACG-motif (**Figure 3a**), as well as a Myb-binding (MBS) and GATA binding site. In addition, *ZmHMA1* contains a binding site for TCP proteins such as Tb1, increasing the number of families of transcription factors that could be involved in shared or individual regulation of these three genes (**Figure 3a**).

To elucidate the expression patterns of *ZmHMA1* and *ZmHMA7*, we conducted reverse transcriptase PCR (RT-PCR) in leaves, stems, roots, and flowers at two vegetative (V2 and V6) and one reproductive (R1) developmental stage, in teosinte *parviglumis* and wild-type W22 maize individuals (**Figure 3b**). In the case of maize, *ZmHMA1* was expressed in leaves, stems, roots

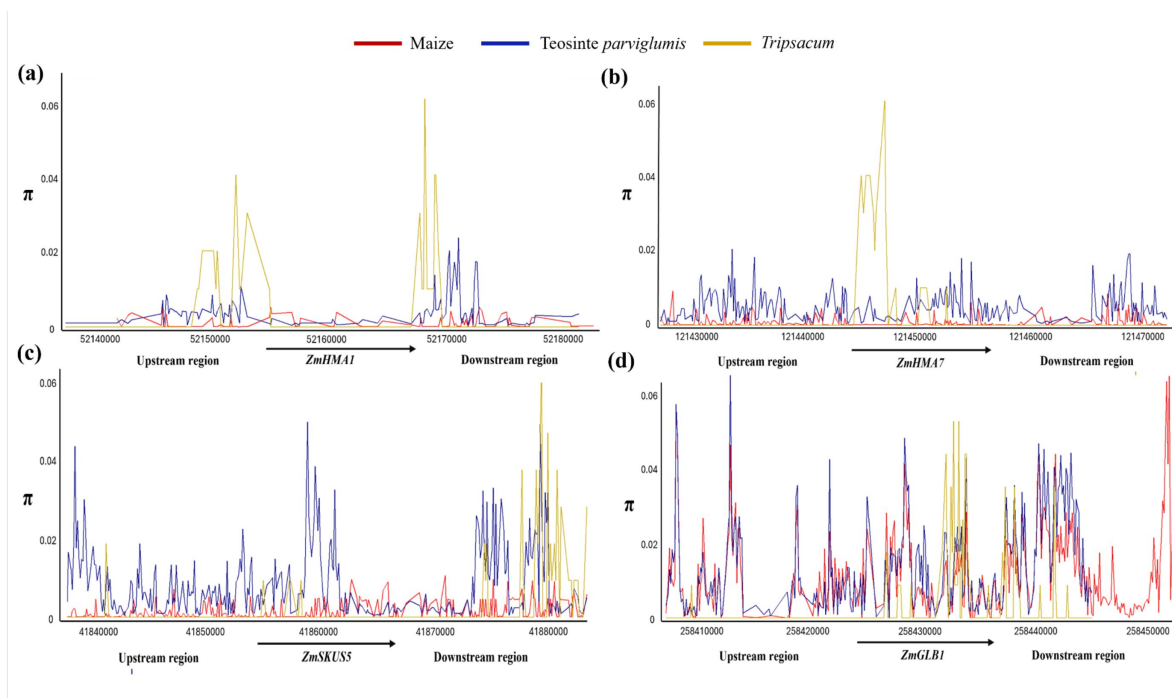


Figure 2.

Genetic diversity of loci encompassing *ZmHMA1*, *ZmHMA7*, and *ZmSKUs5*.

The nucleotide variability index (π) was calculated for all HapMap3 accessions of teosinte *parviglumis* (blue line), maize (red line), and *Tripsacum dactyloides* (orange line), taking in consideration the coding sequence (arrow) and an upstream and downstream region of 15 Kb encompassing each gene. **(a)** Nucleotide variability in the *ZmHMA1* locus. **(b)** Nucleotide variability in the *ZmHMA7* locus. **(c)** Nucleotide variability in the *ZmSKUs5* locus. **(d)** Nucleotide variability in the *ZmGLB1* locus.

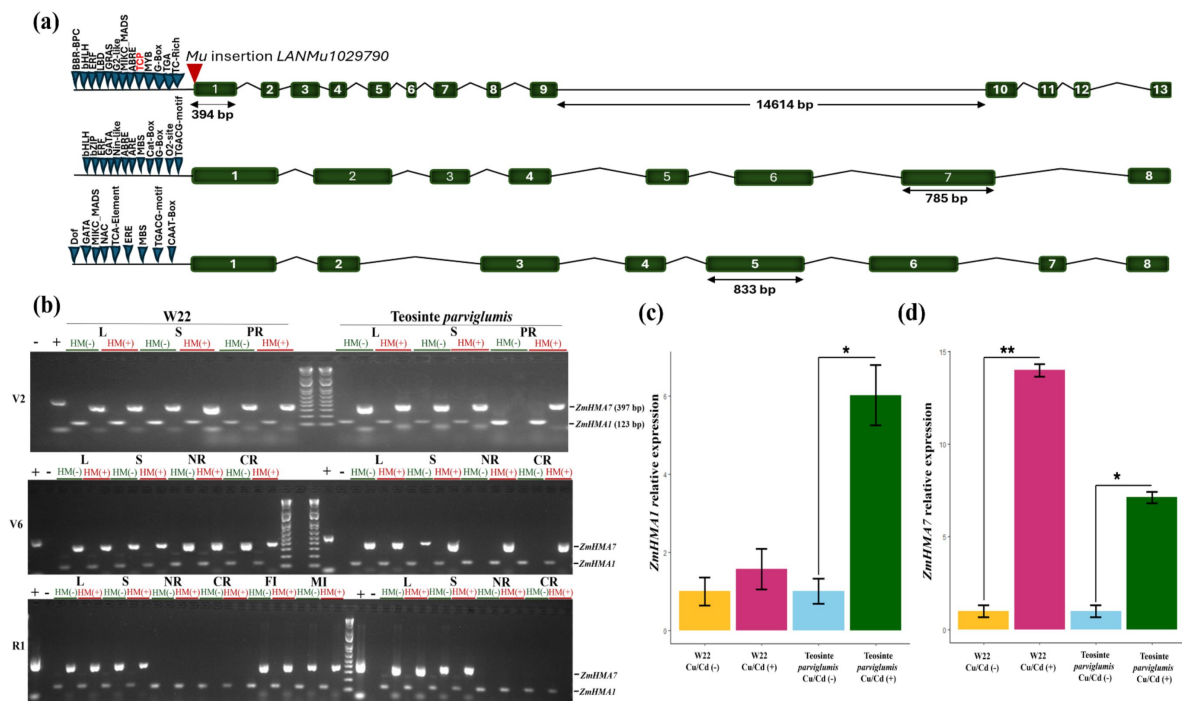


Figure 3.

Structure and gene expression of *ZmHMA1* and *ZmHMA7* in teosinte *parviglumis* and W22 maize.

(a) Structural analysis of *ZmHMA1*, *ZmHMA7*, and *ZMSKU5*. Blue triangles indicate the location of motifs predicted in the regulatory region of each gene. The insertion site of *Mu* line LANMu1029790 is indicated by a red triangle. (b) RT-PCR analysis of *ZmHMA1* and *ZmHMA7* expression throughout development of teosinte *parviglumis* and W22 maize in presence or absence of heavy metal stress; L=leaf; S=stem; PR=primary root; NR=nodal root; CR=crown root. V2: 15 days after transplant; V6: 30 days after transplant; R1: 50 days after transplant. Controls: (-) absence of DNA; (+) genomic DNA. (c) qPCR analysis of *ZmHMA1* expression in 1-month old primary roots, in absence or presence of heavy metal stress. (d) qPCR analysis of *ZmHMA7* expression in 1-month old primary roots, in absence or presence of heavy metal stress. * $P < 0.01$; ** $P < 0.001$.

and flowers throughout development in either absence or presence of HMs. *ZmHMA7* was also expressed in aerial organs throughout development in both absence and presence of HMs but was not expressed in roots at reproductive stage R1. In teosinte *parviglumis*, *ZmHMA1* was also expressed throughout development in either absence or presence of HMs. By contrast, *ZmHMA7* was constitutively expressed in aerial organs, but only expressed in roots under HM stress during V2 and V6 stages. To further determine if the level of expression of these two genes is influenced by the presence of HMs, we conducted quantitative real-time PCR (qPCR) using the primary root of teosinte *parviglumis* and maize plantlets at four weeks after germination (**Figures 3c** and **3d**). In wild-type W22 maize, the expression of *ZmHMA1* was not significantly different in the presence and absence of HMs. By contrast, in teosinte *parviglumis* the expression of *ZmHMA1* was significantly increased in presence of HMs (**Figure 3c**). In the case of *ZmHMA7*, expression was significantly increased in both maize and teosinte *parviglumis* when plants were grown in the presence of HMs (**Figure 3d**). Overall, these results indicate that the activity of both *ZmHMA1* and *ZmHMA7* is influenced by HM stress in teosinte *parviglumis* and maize.

ZmHMA1 null mutant individuals show phenotypic defects related to the domestication syndrome

Transposon *Mu* insertional lines have been widely used to elucidate gene function in maize. To elucidate the function of *ZmHMA1*, we phenotypically analyzed *LANMu1029790*, a 4-generation backcross of line *mu1029790* into the W22 background selected for single copy insertions in *ZmHMA1* (McCarty et al. 2005). In *LANMu1029790*, a single *Mu* insertion is in the first exon of *ZmHMA1*, 65 bp downstream of the START codon (**Figure 3a**). RT-PCR analysis showed that expression of *ZmHMA1* is completely absent in homozygous individuals for the *LANMu1029790* insertion (**Figure S3**), confirming that the corresponding mutant is a null allele. Plants harboring a homozygous *LANMu1029790* insertion are subsequently designated as *zmhma1-m1::MuDR* (abbreviated *zmhma1-m1*).

To determine the function of *ZmHMA1* in maize, we compared the phenotype of wild-type W22 and homozygous *zmhma1-m1* individuals at the reproductive R1 stage (**Table 2**). Under normal growing conditions, homozygous *zmhma1-m1* individuals generated significantly less leaves than wild-type (NL = 13.4 ± 0.84 in wild-type, and NL = 11.4 ± 0.84 in *zmhma1-m1*; $P < 0.001$). Whereas the length of mutant leaves was smaller than wild-type (LML = 77.4 ± 6.64 cm in wild-type, and LML = 69.85 ± 9.18 cm in *zmhma1-m1*), their width was significantly larger (WML = 6.75 ± 1.3 cm in wild-type, and WML = 9.75 ± 1.4 cm in *zmhma1-m1*; $P < 0.001$), resulting in an increase of the total dry weight of aerial organs in mutant plants (DWAP = 18.43 ± 8.04 grams in wild-type, and DWAP = 28.46 ± 9.54 grams in *zmhma1-m1*). Homozygous *zmhma1-m1* individuals also produced significantly more female inflorescences per plant (NFI = 1.2 ± 0.42 in wild-type, and NFI = 1.9 ± 0.32 in *zmhma1-m1*; $P < 0.001$) a difference that reflected in the total dry weight of female inflorescences per plant (WFI = 40.06 ± 14.39 gr in wild-type and WFI = 69.5 ± 11.2 gr in *zmhma1-m1*; $P < 0.001$); and significantly more spikes per male inflorescence (NSB = 9 ± 1.63 in wild-type, and NSB = 13.4 ± 2.32 in *zmhma1-m1*; $P < 0.001$). Interestingly, the general length of roots was significantly smaller in homozygous *zmhma1-m1* individuals, in particular nodal roots (LNR = 87.2 ± 16.8 cm in wild-type, and LNR = 50.9 ± 12.83 cm in *zmhma1-m1*; $P < 0.001$); however, mutant plants showed a significant increase in the number of seminal roots per plant (NSR = 4.1 ± 1.45 in wild-type, and NSR = 7.3 ± 1.16 in *zmhma1-m1*; $P < 0.001$). When wild-type and mutant individuals were grown under HM stress, differences in the size of leaves, number of female inflorescences per plant, and number of spikes per male inflorescence were maintained. In addition, mutant individuals exposed to HM stress were significantly taller than wild-type (HEI = 82.2 ± 34.94 cm in wild-type, and HEI = 125.9 ± 18.81 cm in *zmhma1-m1*; $P < 0.01$). In addition, total root and seminal root length were equivalent in wild-type and mutant individuals, mainly because the length of wild-type roots was significantly reduced under HM stress; however, the number of seminal roots was again significantly higher in mutant individuals (NSR = 6.6 ± 1.35 in wild-type, and NSR = 8.2 ± 1.62 in *zmhma1-m1*; $P < 0.05$). These results indicate that in maize aerial organs, *ZmHMA1* is

involved in promoting leaf growth, restricting the number of female inflorescences per plant, and restricting the number of spikes in male inflorescences. In underground organs, *ZmHMA1* is involved in promoting root growth and restricting the number of seminal roots per individual. Under HM stress, *ZmHMA1* shows an additional role in restricting plant height, a function that was not identified under normal growing conditions. Overall, these results suggest that *ZmHMA1* is involved in the control of some of the aerial and underground phenotypic traits that distinguish maize from teosinte *parviglumis*, especially under HM stress conditions.

Under heavy metal stress, *Tb1* is overexpressed in the apical meristem of teosinte *parviglumis*

The absence of secondary ramifications in teosinte *parviglumis* grown under HM stress is reminiscent of the aerial architecture of wild-type maize. Previous QTL and molecular analysis suggested that *Tb1* contributed to the architectural difference between maize and teosinte, as *tb1* maize mutants develop secondary stem ramifications ending in male inflorescences, resembling teosinte in their overall architecture (Doebley et al. 1997 [DOI](#)). Interestingly, the regulatory region of *ZmHMA1* contains a specific binding site for members of the Teosinte-branched 1/Cycloidea/Proliferating (TCP) family of transcription factors that includes *Tb1* (Figure 3a [DOI](#)). To investigate the possibility that the phenotypic response of teosinte *parviglumis* under HM stress could be regulated by *Tb1*, we conducted quantitative real-time PCR (qPCR) to determine the expression of *ZmHMA1* and *Tb1* in the shoot apical meristem (SAM) of one month old W22 maize and teosinte *parviglumis* plantlets (Figure 4a [DOI](#) and 4b [DOI](#)). The expression *ZmHMA1* was downregulated in W22 wild-type maize meristems in the presence of HMs. By contrast, HM stress resulted in a significant increase of *ZmHMA1* expression in teosinte *parviglumis*, confirming that in the maize ancestor the level of *ZmHMA1* activity is influenced by the presence of HMs in the soil. In addition, the expression of *Tb1* was significantly upregulated in teosinte *parviglumis* meristems in the presence of HMs, suggesting that formation of secondary ramifications in teosinte *parviglumis* is repressed by *Tb1* function under HM stress. Taken together, these results open the possibility for a direct positive regulation of *ZmHMA1* by transcriptions factor of the TCP family, including *Tb1*.

Discussion

Although the evolutionary transition from teosinte *parviglumis* to maize resulted in important phenotypic divergence between landraces and their wild progenitor, the relevance of teosinte *parviglumis* phenotypic plasticity and its genetic consequences during domestication remain poorly understood. Our study offers the first evidence showing that a specific environmental factor (heavy metal stress) influenced specific gene function (heavy metal response ATPases) and human selection during the *parviglumis* to maize transition. The comparison of HM responses between both subspecies indicates that teosinte *parviglumis* phenotypically responds to HM stress acquiring phenotypes reminiscent of maize. Whereas low temperatures and CO₂ concentrations also caused a significant decrease in lateral branching, these environmental response has not been related to specific gene function and human selection. The absence of tillering exhibited by teosinte *parviglumis* under HM stress was not reported under conditions simulating an early Holocene environment (Piperno et al., 2015 [DOI](#)). Our results also show an increase in the number of female inflorescences per plant. This HM stress-dependent proliferation of several closely associated female inflorescences in a single teosinte *parviglumis* stem has not been previously reported. Female inflorescences arose from individual axillary meristems in multiple internodes, significantly increasing teosinte *parviglumis* seed yield. Our evidence indicates that HM stress revealed a teosinte *parviglumis* environmental plasticity that is directly related to the function of specific HM response genes that were affected by domestication through human selection.

Table 2.

Estimation of phenotypic traits in wild-type and *zmhma1* maize grown in absence and presence of heavy metal stress.

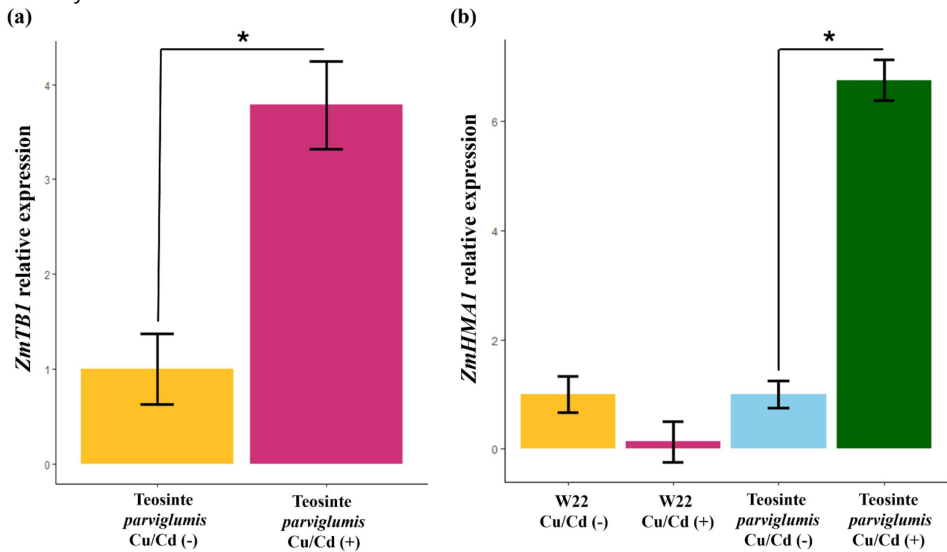
	Description	Cu/Cd(-)					Cu/Cd(+)				
		W22		<i>zmhma1</i>		P value	W22		<i>zmhma1</i>		P value
		Mean	SD	Mean	SD		Mean	SD	Mean	SD	
Aerial traits	NL Number of Leaves	13.40	0.84	11.40	0.84	0.00005***	12.00	1.70	11.10	0.74	0.14194
	LML Length of Mature Leaf (cm)	77.40	6.64	69.85	9.18	0.04930*	72.35	8.81	63.85	7.31	0.03054*
	WML Width of Mature Leaf (cm)	6.75	1.30	9.75	1.40	0.00010***	7.20	1.42	9.95	0.86	0.00006***
	LIL Length of Immature Leaf (cm)	59.50	6.10	48.25	6.88	0.00112**	51.90	9.97	45.35	9.43	0.14868
	WIL Width of Immature Leaf (cm)	7.45	0.80	9.25	0.72	0.00005***	6.60	1.10	8.85	0.85	0.00007***
	HEI Height (cm)	99.60	32.61	103.15	10.76	0.74750	82.20	34.94	125.90	18.81	0.00266**
	SD Stem Diameter (cm)	3.24	0.52	3.02	0.51	0.34945	2.60	0.53	2.86	0.59	0.31207
	ADWAP Dry Weight of Aerial Part (g)	28.43	8.04	28.46	9.54	0.02042*	25.22	8.26	31.44	18.85	0.00156**
	NSB Number of Spike Branches	9.00	1.63	13.40	2.32	0.00011***	7.20	3.16	13.60	1.96	0.00004***
	NFI Number of Female Inflorescences	1.20	0.42	1.90	0.32	0.00054***	0.70	0.67	1.60	0.52	0.00357***
Root traits	WFI Weight of Female Inflorescences (g)	40.06	14.39	69.53	11.20	0.00007***	37.75	18.83	76.06	25.04	0.00113**
	NBR Number of Brace Roots	12.90	4.33	10.60	6.29	0.35370	16.30	4.99	9.50	6.79	0.01998*
	LBR Length of Brace Roots (cm)	16.15	9.75	15.75	14.03	0.94179	4.00	2.04	2.67	2.01	0.15890
	NCR Number of Crown Roots	11.30	2.11	12.10	2.42	0.44153	7.60	0.97	9.00	2.58	0.12569
	LCR Length of Crown Roots (cm)	58.90	21.87	50.50	13.44	0.31442	48.85	11.80	44.40	20.75	0.56281
	NNR Number of Nodal Roots	10.50	2.27	8.50	2.42	0.07262	6.70	2.79	8.00	1.56	0.21505
	LNR Length of Nodal Roots (cm)	87.20	16.80	50.90	12.83	0.00004***	55.30	21.10	53.90	31.59	0.90850
	NSR Number of Seminal Roots	4.10	1.45	7.30	1.16	0.00004***	6.60	1.35	8.20	1.62	0.02743*
	LSR Length of Seminal Roots (cm)	27.70	13.38	21.33	10.31	0.24838	43.50	14.01	53.90	20.85	0.20694

* P<0.05; ** P<0.01; *** P<0.001.

Figure 4.

Tb1 and *ZmHMA1* expression in the shoot apical meristem of plantlets grown under heavy metal stress.

(a) qPCR analysis of *Tb1* expression in 1-month old teosinte *parviglumis* plantlets grown in absence or presence of heavy metal stress. (b) qPCR analysis of *ZmHMA1* expression in 1-month old teosinte *parviglumis* and W22 maize plantlets in absence or presence of heavy metal stress. * P<0.01.



We also show that *ZmHMA1*, *ZmHMA7* and *ZmSKUs5* exhibit reduced genetic diversity in extant maize as compared to teosinte *parviglumis*, suggesting that all three loci were affected by the evolutionary transition that gave rise to maize. The maize regulatory region and downstream 3' area of the *ZmHMA1* locus exhibit pronounced reduction in SNPs compared to teosinte *parviglumis* lines, likely affecting the molecular scope of its regulatory proteins, including transcription factors of the TCP family (Martín-Trillo and Cubas 2010). By contrast, *ZmHMA7* and *ZmSKUs5* showed a selective sweep across the entire maize locus. Although loss of genetic diversity is usually the result of human selection during domestication, it can also represent a consequence of natural selective pressures favoring fitness of specific teosinte *parviglumis* allelic variants better adapted to environmental changes and subsequently affected by human selection during the domestication process. The significant reduction of nucleotide variability in the *ZmHMA1* locus of teosinte *parviglumis* suggests that indeed that a natural selection pressure. Examples of edaphological factors driving genetic divergence in the teosintes or maize include local adaptation to phosphorus concentration in *mexicana* and *parviglumis* (Aguirre-Liguori et al. 2019), and fast maize adaptation to changing iron availability through the action of genes involved in its mobilization, uptake, and transport (Benke and Stich 2011).

The maize genome contains 12 genes encoding for members of the P1b family of ATPases (Cao et al. 2019). They are subdivided according to their preferential substrate affinity: the Copper (Cu)/Silver (Ag) group includes *ZmHMA1* to *ZmHMA4*; and the Zinc (Zn)/Cobalt (Co)/Cadmium (Cd)/Lead (Pb) includes *ZmHMA5* to *ZmHMA12*. Initial studies suggested that *ZmHMA1* is expressed at low levels in vegetative, reproductive or root organs throughout development, contrary to *ZmHMA7* that is strongly expressed in developing seeds 10 days after pollination; however, both genes appeared downregulated in plants grown under either Cu or Cd stress (Zhiguo et al. 2018). Additional studies by Cao et al (2019) indicated low expression of *ZmHMA1* and high levels of *ZmHMA7* expression in 12 tissues: roots, nodes, shoots, leaves, tassel, shoot apical meristem and young stem, internodes, anthers, silks, whole seedlings, endosperm, embryo, and pericarp. Under Cd stress, expression of both *ZmHMA1* and *ZmHMA7* was significantly upregulated in roots and downregulated in stems (Cao et al. 2019), contradicting previous results by Zhiguo et al. (2018). In Arabidopsis and barley, the corresponding *ZmHMA1* structural homologs (*AtHMA1* and *HvHMA1*, respectively) can transport a wide range of metals into the chloroplast (Mikkelsen et al. 2012; Boutigny et al. 2014). In the case of the Arabidopsis structural homolog of *ZmHMA7*, *AtHMA7* (also called RAN1) is proposed to act as a metallochaperone located in the Golgi apparatus, delivering Cu to the ethylene receptor located in the plasma membrane (Hirayama et al. 1999). Our evidence complements previous studies by showing that, in maize, *ZmHMA1* root expression is restricted to vegetative developmental stages, and that *ZmHMA7* is expressed in roots in response to HM stress. We also show through qPCR analysis that, under HM stress, the expression of both *ZmHMA1* and *ZmHMA7* is upregulated in teosinte *parviglumis* primary roots of 4-week-old plantlets. Overall, these results confirm that both *ZmHMA1* and *ZmHMA7* are heavy metal response genes active throughout teosinte and maize development.

In maize, the phenotypic analysis of null mutant individuals indicates that *ZmHMA1* is involved in the control of aerial and underground developmental traits, some of which are related to important functional adaptations that distinguish maize from its wild ancestor. In addition to changes in leaf number, leaf length, and number of spikes per male inflorescence, *zmhma1-m1* mutant individuals grown under normal edaphological conditions showed a significant increase in the number of female inflorescences per plant, a significant reduction in the length of nodal roots, and a significant increase in the number of seminal roots per plant. Whereas the promotion of nodal root growth is a distinctive trait of maize phenotypic plasticity and adaptation, previous studies showed that contrary to maize, plants of teosinte *parviglumis* lack seminal roots. This absence is also characteristic of paleobotanical maize specimens dating more than 5,000 years before present, confirming that the acquisition of seminal roots is an important trait acquired during domestication (Lopez-Valdivia et al. 2022). The presence of seminal roots distinguishes

extant maize from the teosintes (Burton et al. 2013 [↗](#)), suggesting a possible involvement in domestication (Perkins and Lynch 2021 [↗](#); Lopez-Valdivia et al. 2022 [↗](#)). Our results suggest that the allelic variants of the *ZmHMA1* locus selected during domestication could have contributed to promote seminal root formation in maize, optimizing their number in accordance to seed endosperm availability and carbohydrate reserves (Perkins and Lynch 2021 [↗](#); Lopez-Valdivia et al. 2022 [↗](#)). Interestingly, *ZmHMA1* plays an important role in contributing to restricting plant height in response to HM stress, a phenotype that is also an important distinction between teosinte *parviglumis* and maize. Taken together, these results indicate that *ZmHMA1* function was involved in the phenotypic transition from teosinte *parviglumis* to maize through the influence of HM stress; however, a better definition of its role in the origin and domestication of maize will require a functional analysis of the *ZmHMA1* teosinte *parviglumis* homolog, and a detailed molecular analysis of the corresponding allelic variants and their phenotypic effects in both subspecies.

Ideas and Speculation

Our results point to an evolutionary model in which environmental changes in the form of edaphological HM stress imposed natural selective forces that acted on native teosinte *parviglumis* populations, causing phenotypic responses and a severe genetic bottleneck (**Figure S4** [↗](#)). Adapted plants developed phenotypic traits such as shorter height and emergence of seminal roots, progressively giving rise to primitive maize individuals with reduced genetic variation in HM response genes such as *ZmHMA1*. The subsequent action of human selection caused a drastic reduction in allelic variation of heavy metal response genes, and genetic fixation of the resulting domestication traits.

The most important source of HMs in ancient soils of Mesoamerica is volcanic activity through short- and long-term effects related to ores, hydrothermal water flow and ash deposits. The history of eruptive activity in the Trans Mexican Volcanic Belt during the early Holocene coincides with the temporal estimation of initial stages in the evolutionary transition that gave rise to maize. Intense and recurrent eruptions occurred 24,000 to 6,000 years ago near the region of the Balsas River basin presumed to be the cradle of maize domestication (Matsuoka et al. 2002 [↗](#); Macias 2005). Located less than 100 Km from the region that harbors the natural teosinte *parviglumis* populations most closely related to maize (Matsuoka et al. 2002 [↗](#)), the Nevado de Toluca volcano underwent four major eruptions approximately 24100, 12100, 10500, and 8500 years before present (yr BP; Macias 2005), causing in the region a period of climatic instability that lasted until 8500 yr BP (Ludlow-Wiechers et al., 2005 [↗](#)). Could volcanic eruptions during the early Holocene have modified soil composition and influence the initial evolutionary transition from teosinte *parviglumis* to primitive maize? The possibility of environmental cues influencing maize emergence was mentioned by Benz and Iltis (1992) [↗](#), but not in relation to volcanic activity. Deciphering the ancient DNA sequence of HM response genes such as *ZmHMA1* in paleobotanical maize specimens found in Central Mexico could help determine if its genetic diversity was already reduced by 5,300 to 6,200 yr BP (Ramos-Madrigal et al., 2016 [↗](#); Vallebuena-Estrada et al., 2016 [↗](#); Benz, 2001 [↗](#)). Our results open the possibility for environmental changes caused by volcanic activity acting as a driving force in the selection of loci involved in adaptive responses such as heavy metal tolerance.

Materials and Methods

Plant materials and growth conditions

Seeds from maize inbred line W22, teosinte *parviglumis* accession CIMMYTMA 29791, and insertional line *LANMu1029790* were sterilized using a 1:1 distilled water and bleach solution, rinsed three times in water, and germinated in MS solid medium. One week-old seedlings were transplanted into PVC tubes (length: 90 cm; diameter: 20 cm) or 20-liter pots filled with a pre-sterilized mix composed of 30% peat moss, 10% vermiculite, 10% perlite, and 50% ground litter,

complemented with 15-9-12 (N-P-K) Osmocote™ fertilizer. For HM growth conditions, 16 mg/kg of cadmium chloride (CdCl_2) and 400 mg/kg of copper sulfate (CuSO_4) were added to the substrate before seedling transplant. All plants were grown under glasshouse conditions at 28°C during day and 24°C during night, with relative humidity comprised between 40 and 50%.

Phenotypic analysis

All plants were harvested at 15 days (V2 stage), 30 days (V6 stage), and 50 days (R1 stage) after transplant. No less than ten plants were measured for each trait and stage. The width and length of immature leaves was measured in the 5th leaf. The length and width of mature leaves in the 3rd leaf. The total number of leaves, total number of stem internodes, plant height, basal stem diameter, total number of stem ramifications, and total fresh and dry weight of the aerial biomass were estimated at the R1 stage. Estimated root traits included the number and length of brace roots (aerial roots), the number and length of crown roots (underground roots originating within 2 cm below the basal node), number and length of nodal roots (underground roots originating more than 2 cm below the basal node), the number and length of seminal roots and maximum length of the root system.

Genetic diversity analysis

The SNP collection of maize chromosome 5 was analyzed on the basis of the HapMap V3.2.1 dataset (Bukowski et al. 2018 [\[1\]](#)), selecting individuals from maize landraces (LR) (Table S6 [\[2\]](#)) and teosinte *parviglumis* inbred line accessions (Teo; Table S7 [\[3\]](#)). A sliding window diversity (π) was calculated using VCFtools v0.1.13 (Danecek et al. 2011 [\[4\]](#)) with a window size of 100 bp (--window-pi 100) and a sliding step of 50 bp (--window-pi-step 50). Regions under selection were extracted from a list of 304 genes with domestication-related selection signatures across the whole chromosome (Hufford et al. 2012 [\[5\]](#); Tables S8), for a total of 385,829 sliding windows (Table S9 and S10). A second set of neutral gene regions comprising 2,373,010 windows was also extracted across the entirety of chromosome 5. These two collections were compared to *ZmHMA1* (GRMZM2G067853), *ZmHMA7* (GRMZM2G029951), and *ZmSKUs5* (GRMZM2G076985), encompassing 2,743 windows (Table S11 and S12). Additionally, we analyzed genetic diversity in the neighboring regions of these candidate genes, which included six genes (*GRMZM2G701784*, *GRMZM2G014508*, *GRMZM2G367857*, *GRMZM2G375607*, *GRMZM2G374375*, and *GRMZM2G067883*), totaling 3,088 windows. We employed a custom Perl script (Vallebuena GitHub repository) to calculate two diversity metrics: fold change ($\text{FC} = \pi(\text{LR})/\pi(\text{Teo})$) and the proportional diversity ($\text{FP} = \pi(\text{LR})/(\pi(\text{LR}) + \pi(\text{Teo}))$). Data visualization was performed using violin plots generated in R v4.2.2.

For an arbitrary locus encompassing 30 Kbs up and downstream of the coding sequence of *ZmMA1*, *ZmHMA7*, and *ZmSKUs5*, the number of segregating sites (S), the number of unique sequences (haplotypes, h), and the nucleotide diversity index (θ) were estimated using DnaSP version 6.0 for all accessions included in HapMap3 (Rozas et al. 2017 [\[6\]](#); Bukowski et al. 2018 [\[7\]](#)). DnaSP was also used to estimate Tajima's D, Fu and Li's D, and Fu and Li's F indexes (Tajima 1989 [\[8\]](#); Fu and Li 1993 [\[9\]](#)). The average proportion of pairwise nucleotide differences per nucleotide site (π) was calculated with VCFtools using bins of 100 bp and steps of 25 bp (Tajima 1983 [\[10\]](#); Weir and Cockerham 1984 [\[11\]](#)). Null hypothesis of equality of parameters followed two-tailed tests as previously described (Rozas et al. 2017 [\[6\]](#)).

Gene expression analysis

Leaves, stems, tassels, female inflorescences, as well as primary, nodal, and crown roots were manually isolated, frozen in liquid nitrogen and stored at 4°C. Total RNA was extracted using TRIZOL (Thermo Fisher Scientific, USA). For reverse-transcriptase PCR (RT-PCR), RNA was converted to coding DNA using the One-step RT-PCR kit (QIAGEN, USA). The full list of primers is presented in Table S8. *ZmCDK* was used as a positive control. PCR amplification was conducted

using the following parameters: at 94°C for 5 min.; 94°C 30 sec., 60°C 30 sec., 72°C 1 min. for 35 cycles; final stop after 72°C for 10 min. For quantitative real-time PCR (qPCR), total RNA was isolated from 4-week-old meristems using TRIZOL. Complementary DNA (cDNA) was synthesized using 4 µg of total RNA, 10 mM oligodTs and Superscript reverse transcriptase II (Invitrogen, USA). Primers for PCR were designed using the online program Primer 3 Plus (v.0.4.0) and verified with Oligocalculator (Sigma Chemical, St Louis MO) to discard dimer formation. PCR efficiencies of target and reference gene were determined by generating standard curves based on serial dilutions prepared for cDNA templates. PCR efficiency was calculated according to the slope of the standard curve (primers with 100% efficiency, the fold equals 2). Each quantitative real-time PCR (qPCR) reaction was performed in a 10 µl volume consisting of 5 µl of 23 SYBR Green PCR reaction Mix (Applied Biosystems, Foster City, CA). 3.5 µl of the DNA template (100 ng/ml), 0.5 µl of the forward primer (5 mM), 0.5 µl of the reverse primer (5 mM), and 0.5 µl of ultrapure water. The qPCR reactions were performed using the CFX96 Touch Real-Time PCR detection System and the data were analyzed using the Bio-Rad CFX Manager software v3.1. The thermal profile consisted of 10 min at 95°C, 40 cycles of 15 sec at 95°C, and 1 min at 60°C. Amplification results were collected at the end of the extension step. Primer sequences for qPCR amplification are listed in **Table S7** [↗](#). A comparative 2-ΔΔCT method was used for determining a relative target quantity of gene expression (Scheffe et al. 2006 [↗](#)), and the cyclin-dependent kinase gene *ZmCDK* (GRMZM2G149286) was used as a control (Hu et al. 2019 [↗](#)). Reproducibility of the results was evaluated for each sample by running three technical and three biological replicates of each of the reactions and each sample.

Statistical Analysis

Statistical analyses were conducted using RStudio, version 4.3.0 (Horton and Kleinman 2015 [↗](#)). The data are presented as mean ± standard deviation and were tested by a paired Student's T-Test. When the effects were significant according to T-Test, the treatments were compared with the Welch two sample T-Test at $P < 0.05$.

Legend of Appendix and Supplementary Files

Appendix

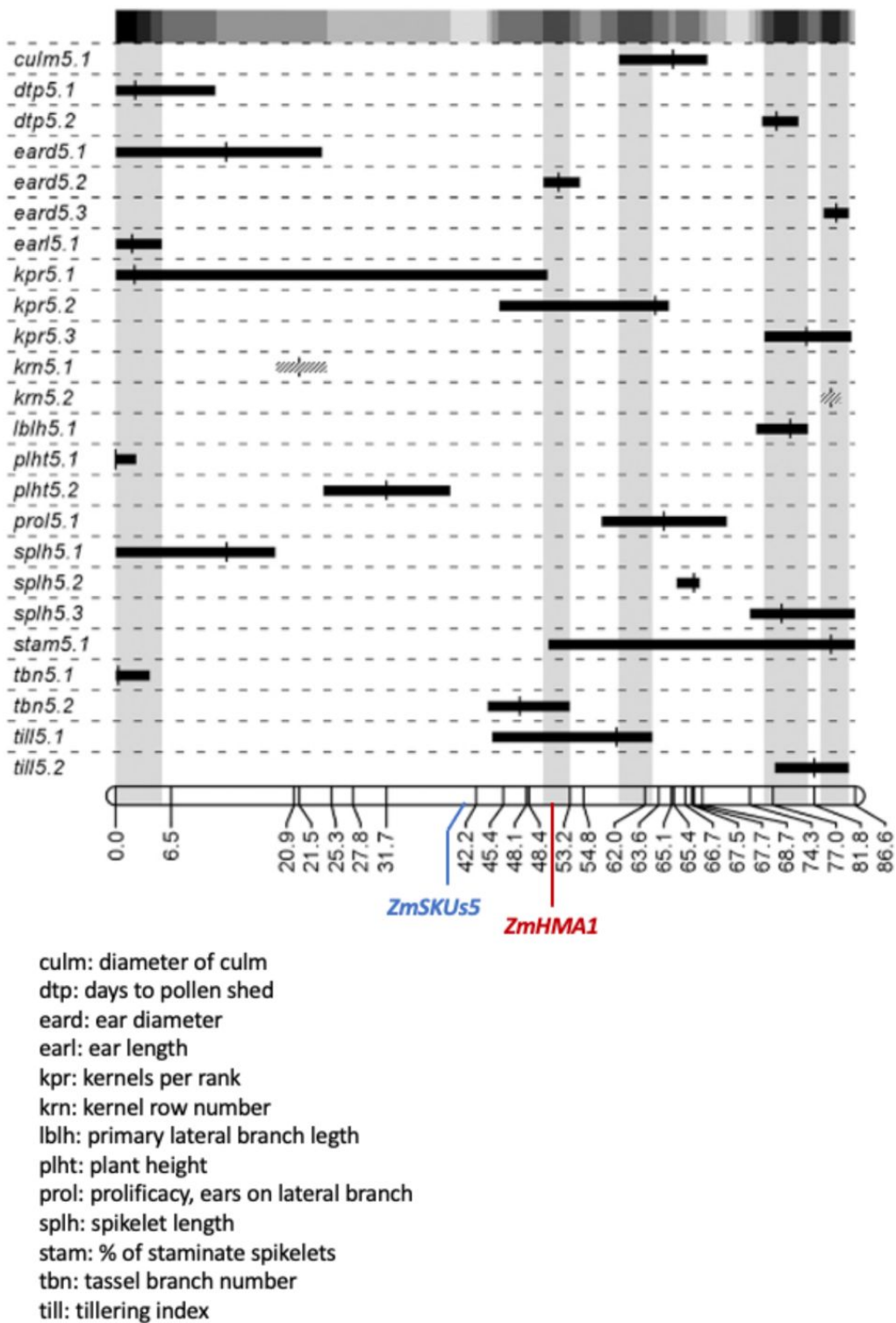


Figure S1.

Cumulative plot of QTL detected in a region of chromosome five that includes *ZmSKUs5* and *ZmHMA1* (modified from Lemmon and Doebley, 2024).

Molecular marker positions and gene location of *ZmSKUs5* and *ZmHMA1* are in centimorgans. Abbreviated QTLs are in the left margin. The 1.5-LOD support intervals for QTLs are in black horizontal bars and peak LOD scores in vertical lines. Hatched bars indicate interacting QTLs. Gray vertical shadings indicate the presence of a QTL cluster.

Fig. S2

Comparison of chromosome 5 nucleotide variability between maize landraces and teosinte *parviglumis*.

Violin plots represent all neutral regions comprised chromosome 5 (blue), domestication genes previously reported in chromosome 5 (Hufford et al., 2012 [\[1\]](#); green), the three candidate genes in this study (white), and their six neighboring genes (red). ANOVA-Tukey tests are represented by letters.

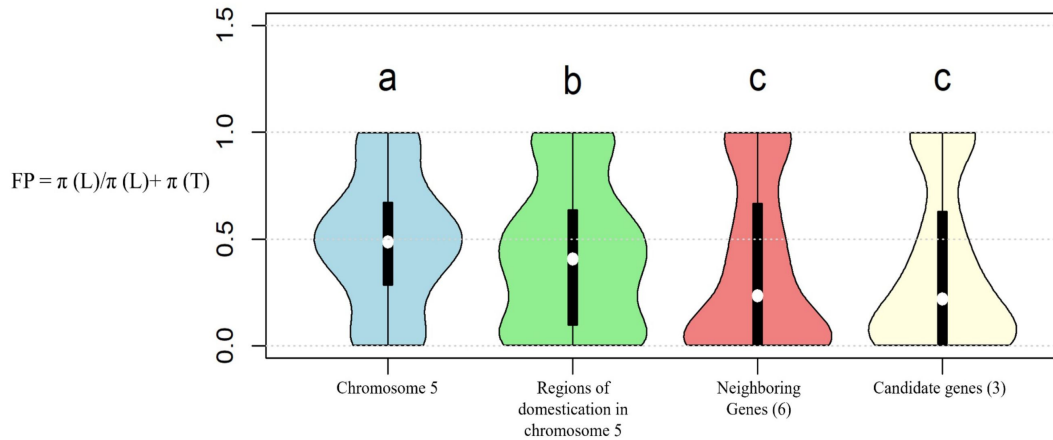
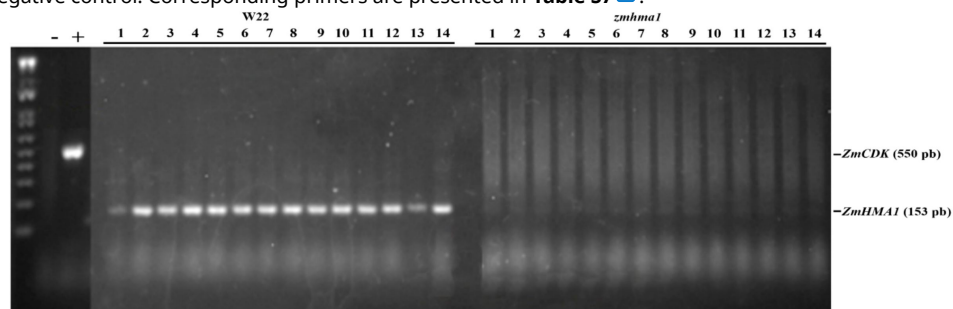


Fig. S3

Expression of *ZmHMA1* in wild-type and homozygous *zmhma1* maize individuals.

A fragment of *ZmHMA1* was the target of RT-PCR amplification in fourteen W22 wild-type and fourteen *zmhma1* homozygous individuals. *ZmCDK* was used as a positive control in Wild-type individual number 1. A reaction without cDNA addition was used as a negative control. Corresponding primers are presented in [Table S7](#) [\[1\]](#).



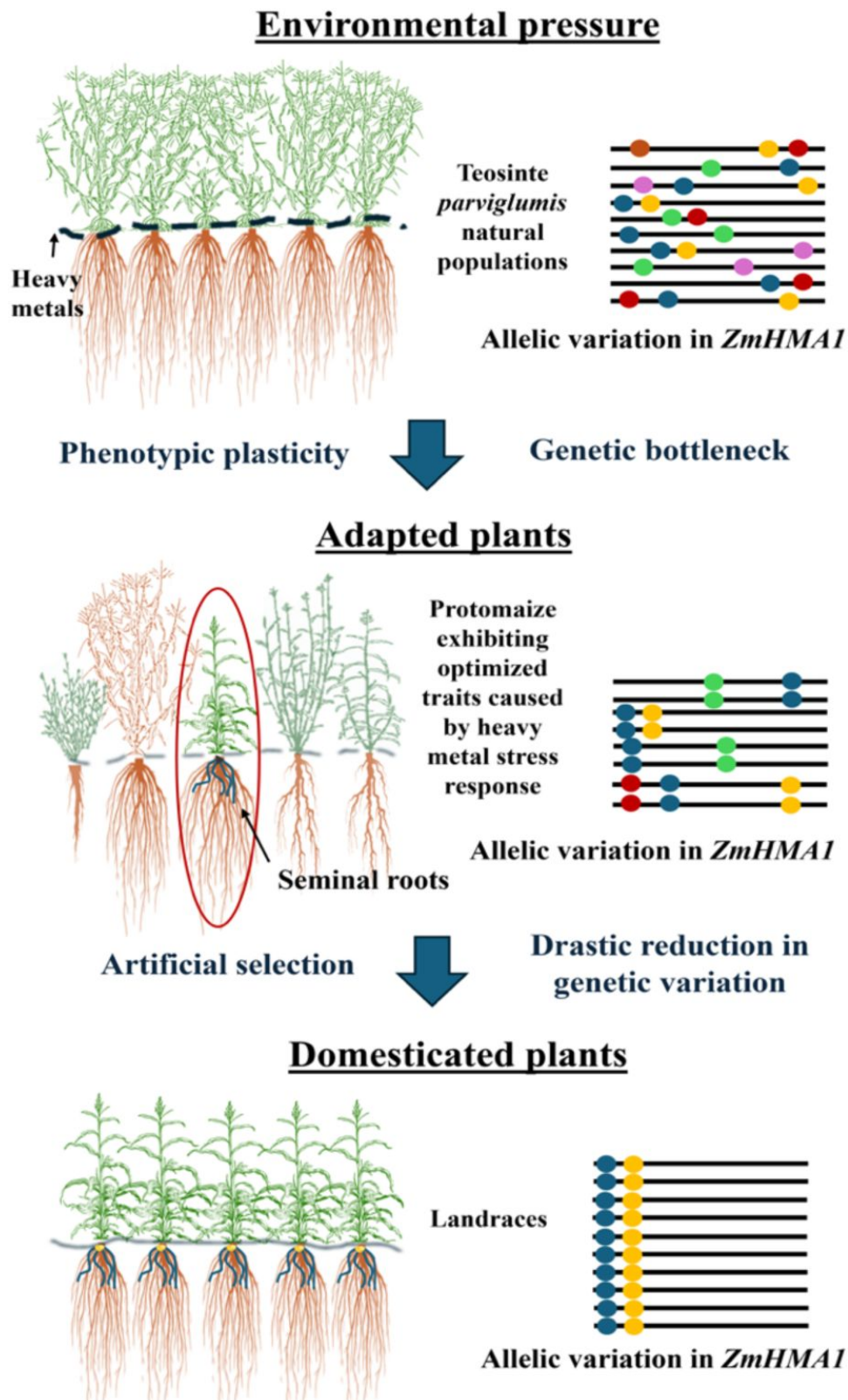


Figure S4.

Model illustrating the influence of heavy metal stress on the evolutionary transition of teosinte *parviglumis* to maize.

Table S1

Additional estimation of phenotypic traits in teosinte *parviglumis* grown in absence or presence of heavy metal stress.

* P<0.05; ** P<0.01.

	Description	Cu/Cd(-)		Cu/Cd(+)		P Value
		Mean	SD	Mean	SD	
Aerial traits	NL Number of Leaves	11.00	2.21	11.40	1.71	0.0360*
	LML Length of Mature Leaf	84.50	4.74	81.30	8.33	0.1662
	WML Width of Mature Leaf	5.45	0.50	6.00	1.25	0.4685
	WIL Width of Immature Leaf	7.10	1.02	6.90	0.61	0.0650
	NN Number of Nodes	9.30	0.95	9.80	1.55	0.0770
	HEI Height	127.30	27.33	134.56	18.51	0.2648
	SD Stem Diameter	2.94	0.66	3.12	0.46	0.0130**
	FWAP Fresh Weight of Aerial Part	667.04	141.02	358.94	49.84	0.7397
	NSB Number of Spike Branches	22.10	4.75	24.70	4.90	0.1323
Root traits	LLBS Length of Longest Branch in Spike	21.25	4.57	20.80	5.95	0.3917
	TRL Total Root Length	105.10	7.10	94.34	44.47	0.3021
	FWR Fresh Weight of Complete Root System	244.60	101.99	225.70	40.05	0.0892
	DWR Dry Weight of Complete Root System	34.21	14.29	31.84	3.57	0.0892

Table S2

Additional estimation of phenotypic traits in wild-type and *zmhma1* maize grown in absence or presence of heavy metal stress.

	Description	Cu/Cd(-)						Cu/Cd(+)					
		W22		<i>zmhma1</i>				W22		<i>zmhma1</i>			
		Mean	SD	Mean	SD	P value		Mean	SD	Mean	SD	P value	
Aerial traits	NN Number of Nodes	11.00	2.36	10.50	1.35	0.56800		9.10	2.18	10.10	1.52	0.25038	
	FWAP Fresh Weight of Aerial Part	242.56	105.73	218.93	73.36	0.56867		226.39	108.55	318.78	145.03	0.12420	
	LLBS Length of Longest Branch in Spike	21.60	1.66	17.85	4.59	0.02596*		20.45	3.74	15.75	6.99	0.07720	
	LFI Length of Female Inflorescences	24.20	2.97	24.10	3.38	0.94479		21.50	3.17	25.50	4.60	0.03618*	
Root traits	TRL Total Root Length	91.90	17.12	72.29	9.51	0.00535**		83.98	16.19	82.33	17.27	0.82821	
	FWR Fresh Weight of Complete Root System	162.20	76.45	78.76	26.24	0.00431**		104.14	47.28	79.85	31.95	0.19497	
	DWR Dry Weight of Complete Root System	6.81	3.21	5.51	1.84	0.28136		4.37	1.98	5.59	2.24	0.21342	

P<0.05; ** P<0.01

Table S3

Full comparison of phenotypic values for wild-type W22, *zmhma1*, and teosinte *parviglumis* individuals grown under absence or presence of heavy metal stress.

	Description	W22				<i>zmhma1</i>				Teosinte <i>parviglumis</i>			
		Cu/Cd(-)		Cu/Cd(+)		Cu/Cd(-)		Cu/Cd(+)		Cu/Cd(-)		Cu/Cd(+)	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Aerial traits	NL	13.40	0.84	12.00	1.70	11.40	0.84	11.10	0.74	11.00	2.21	11.40	1.71
	LML	77.40	6.64	72.35	8.81	69.85	9.18	63.85	7.31	84.50	4.74	81.30	8.33
	WML	6.75	1.30	7.20	1.42	9.75	1.40	9.95	0.86	5.45	0.50	6.00	1.25
	LIL	59.50	6.10	51.90	9.97	48.25	6.88	45.35	9.43	71.30	11.92	56.25	7.80
	WIL	7.45	0.80	6.60	1.10	9.25	0.72	8.85	0.85	7.10	1.02	6.90	0.61
	NN	11.00	2.36	9.10	2.18	10.50	1.35	10.10	1.52	9.30	0.95	9.80	1.55
	HEI	99.60	32.61	82.20	34.94	103.15	10.76	125.90	18.81	127.30	27.33	134.56	18.51
	SD	3.24	0.52	2.60	0.53	3.02	0.51	2.86	0.59	2.94	0.66	3.12	0.46
	SB	ND	ND	ND	ND	ND	ND	ND	ND	5.30	1.83	0.60	0.84
	FWAP	242.56	105.73	226.39	108.55	218.93	73.36	318.78	145.03	667.04	141.02	358.94	49.84
	DWAP	18.43	8.04	17.22	8.26	28.46	9.54	41.44	18.85	133.49	28.28	82.72	11.67
	NSB	9.00	1.63	7.20	3.16	13.40	2.32	13.60	1.96	22.10	4.75	24.70	4.90
	LLBS	21.60	1.66	20.45	3.74	17.85	4.59	15.75	6.99	21.25	4.57	20.80	5.95
	NFI	1.20	0.42	0.70	0.67	1.90	0.32	1.60	0.52	1.80	0.63	3.90	1.66
	LFI	24.20	2.97	21.50	3.17	24.10	3.38	25.50	4.60	15.40	1.35	12.50	2.12
	WFI	40.06	14.39	37.75	18.83	69.53	11.20	76.06	25.04	1.29	0.74	1.29	0.56
Root traits	NBR	12.90	4.33	16.30	4.99	10.60	6.29	9.50	6.79	10.60	6.13	11.60	4.55
	LBR	16.15	9.75	4.00	2.04	15.75	14.03	2.67	2.01	20.00	11.49	2.88	0.66
	NCR	11.30	2.11	7.60	0.97	12.10	2.42	9.00	2.58	12.80	4.34	9.60	1.43
	LCR	58.90	21.87	48.85	11.80	50.50	13.44	44.40	20.75	60.70	16.95	62.80	14.10
	NNR	10.50	2.27	6.70	2.79	8.50	2.42	8.00	1.56	16.20	4.69	14.00	3.20
	LNR	87.20	16.80	55.30	21.10	50.90	12.83	53.90	31.59	83.00	15.76	104.00	8.46
	NSR	4.10	1.45	6.60	1.35	7.30	1.16	8.20	1.62	ND	ND	ND	ND
	LSR	27.70	13.38	43.50	14.01	21.33	10.31	53.90	20.85	ND	ND	ND	ND
	TRL	91.90	17.12	83.98	16.19	72.29	9.51	82.33	17.27	105.10	7.10	94.34	44.47
	FWR	162.20	76.45	104.14	47.28	78.76	26.24	79.85	31.95	244.60	101.99	225.70	40.05
	DWR	6.81	3.21	4.37	1.98	5.51	1.84	5.59	2.24	34.21	14.29	31.84	3.57

Table S4

Nucleotide variability across the *ZmSKUS5*, *ZmHMA1* and *ZmHMA7* locus; and their upstream and downstream neighboring genes in maize landraces, maize improved lines, and teosinte *parviglumis* accessions included in HapMap3.

Gene ID	GeneLength(Kp)	DistanceOfGene(Kp)	Upstream(30Kb)			Coding Region			Downstream(30Kb)		
			π (Maize)	π (Teosinte)	π M/ π T	π (Maize)	π (Teosinte)	π M/ π T	π (Maize)	π (Teosinte)	π M/ π T
GRMZM2G375607 ^a	0.17	37.3	0.00105	0.000675	1.5556	0.037894	0.0304718	1.2436	0.00041	0.00064	0.6406
GRMZM2G076985 ^b	4.17	-	0.011884529	0.017749593	0.669566282	0.000437979	0.007541557	0.058075436	0.002030405	0.006264344	0.324120984
GRMZM2G014508 ^c	22.8	97.2	0.00844561	0.005282053	1.598925645	0.037149063	0.011257665	3.299890491	0.002433718	0.017648094	0.13790262
GRMZM2G367857 ^d	8.72	25.7	0.005243813	0.006394847	0.820006014	0.008991137	0.011367691	0.790937843	0.002425106	0.006884682	0.352246578
GRMZM2G067853 ^e	21.6	-	0.01329488	0.01772011	0.750270738	0.005748	0.0010578	5.433919455	0.00096874	0.0024874	0.389458873
GRMZM2G067883 ^f	0.78	24.9	0.005048118	0.007825611	0.645076531	0.00119172	0.000102418	11.63584526	0.003030221	0.00681471	0.444658834
GRMZM2G374375 ^g	1.84	228.6	0.0007481	0.000974	0.768069815	0.000974	0.01495	0.651505017	0.00054	0.00948	0.056962025
GRMZM2G029951 ^h	4.5	-	0.000678	0.010784	0.06287092	0.00015	0.00148	0.101351351	0.00066	0.00454	0.145374449
GRMZM2G701784 ⁱ	0.54	56	0.0005974	0.009748	0.061284366	0.00074	0.01784	0.041479821	0.00017	0.00009	1.888888889
GRMZM2G067919 ^j	2.4	-	0.00611	0.00809	0.755253399	0.01035	0.01058	0.97826087	0.01417	0.01254	1.129984051

^aUncharacterized gene; ^b*ZmSKUS5* (Multicopper oxidase); ^c*ZmDds_sub1* (Decaprenyl-diphosphate synthase subunit 1);

^dUncharacterized gene; ^e*ZmHMA1*; ^fUncharacterized gene; ^g*Zmpgl20* (Exopolysaccharuronase-like); ^h*ZmHMA7*; ⁱUncharacterized gene;

^j*ZmGLB1* (Globulin1)

Table S5.

Collection of primers used in this study.

Primer Name	Gene ID	Forward	Reverse
<i>ZmHMA1</i> _PCR/qPCR	GRMZM2G067853_T02	TGAACACTCCGACATGGTCC	GGTGTAGCTTGAGGGAGAGC
<i>ZmHMA7</i> _PCR/qPCR	GRMZM2G029951_T01	ATCTCCTGCGCTTCTTGTGC	CCTCTGGCCTGTACTGAACC
<i>ZmTb1</i> _qPCR	Zm00001eb054440	CCCATGGACTTACCGCTTA	GGGAGCATGGGTAGCAGTAG
<i>ZmCDK</i> _PCR	Zm00001eb215750_T001	GGAAGGTATGCACAGGACAGAT	TTCAGCACAATCTTGGCAAAAC
<i>ZmCDK</i> _qPCR	Zm00001eb215750_T001	CCGTCATCGCCTCACGAAGAG	AGAGCCTGCCTTACGGAATTGG
<i>ZmHMA1</i> genotyping	GRMZM2G067853_T02	GACGACACAAGCCAAAACAG	AGGGAGATGCAGCAGAGGT

Table S6.

Maize landrace accessions used in the genetic diversity analysis of chr5.

1	BKN009
2	BKN010
3	BKN011
4	BKN014
5	BKN015
6	BKN016
7	BKN017
8	BKN018
9	BKN019
10	BKN020
11	BKN022
12	BKN023
13	BKN025
14	BKN026
15	BKN027
16	BKN029
17	BKN030
18	BKN031
19	BKN032
20	BKN033
21	BKN034
22	BKN035
23	BKN040

1	TIL01
2	TIL01-JD
3	TIL02
4	TIL03
5	TIL04-TIP285
6	TIL04-TIP454
7	TIL05
8	TIL06-TIP260
9	TIL06-TIP496
10	TIL07
11	TIL09
12	TIL10
13	TIL11
14	TIL12
15	TIL14-TIP498
16	TIL15
17	TIL16
18	TIL17

Table S7.

Teosinte *parviglumis* accessions used in the genetic diversity analysis of chr5.

Acknowledgements

We would like to thank Ueli Grossniklaus for critical comments of this manuscript; Juana de la Cruz for supporting greenhouse activities; Ivan López-Valdivia, Eduardo González-Orozco and Cristian Eduardo Martínez Guerrero for help with bioinformatic analysis; and Enrique Pola for help with qPCR analysis. All members of the Apolab offered valuable suggestions and support. This research was funded by Conahcyt grant (CB-256826) and the INAH-Cinvestav Tehuacán Collaborative Initiative. J.A.-B. is the recipient of a graduate scholarship from Consejo Nacional de Humanidades Ciencia y Tecnología (CONAHCyT).

Additional information

Author contributions

J. A-B and J.P. V-C designed the study. J. A-B and J.P. V-C coordinated and managed the project. M.V.E. designed and performed computational analysis. J. gfrtcdA-B performed the experiments, J. A-B, M.V.E and J.P. V-C analyzed and interpreted the data, J. A-B and J.P. V-C wrote the article. All authors read and approved the final version of the manuscript.

Additional files

Supplementary File 1. List of domestication genes present on chr5 as defined by Hufford et al. 2012. [↗](#)

Supplementary File 2. FC and FP values for all domestication genes present in chr5. [↗](#)

Supplementary File 3. FC and FP values of neutral regions included in chr5. [↗](#)

Supplementary File 4. FC and FP values of candidate genes ZmHMA1, ZmHMA7 and ZmSKUs5. [↗](#)

Supplementary File 5. FC and FP values of ZmHMA1, ZmHMA7 and ZmSKUs5 neighboring genes. [↗](#)

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Reviewer #1 (Public review):

In this study, Acosta-Bayona et al. aim to better understand how environmental conditions could have influenced specific gene functions that may have been selected for during the domestication of teosinte parviglumis into domesticated maize. The authors are particularly interested in identifying the initial phenotypic changes that led to the original divergence of these two subspecies. They selected heavy metal (HM) stress as the condition to investigate. While the justification for this choice remains speculative, paleoenvironmental data would add value; the authors hypothesize that volcanic activity near the region of origin could have played a role.

The authors exposed both maize and teosinte parviglumis to a fixed dose of copper and cadmium, representing an essential and a non-essential element, respectively. They assessed shoot and root phenotypic traits at a defined developmental stage in plants exposed to HM stress versus controls. They then focused on three genes already known to help plants manage HM stress: ZmHMA1, ZmHMA7, and ZmSKUs5. Two of these genes are located in a genomic region linked to traits selected during domestication. A closer examination of nucleotide variability in the coding and flanking regions of these genes provided evidence of selective pressure among teosinte parviglumis, maize, and the outgroup *Tripsacum dactyloides*.

They further generated a null mutant for ZmHMA1 and showed, for the first time in maize, a pleiotropic phenotype reminiscent of traits associated with the domestication syndrome. Finally, using qPCR, they reported increased expression of the domestication gene *Teosinte branched1 (tb1)* in teosinte parviglumis under HM stress. Comparative studies focusing on teosinte parviglumis and the genes ZmHMA1, ZmHMA7, and ZmSKUs5 under HM stress are limited; thus, this phenotypic characterization provides a promising starting point for further understanding the genetic basis of the response.

The dataset is of good quality, but the conclusions are not sufficiently supported by the data. Analyses should be expanded, and additional experiments included to strengthen the findings.

(1) Although the paper presents some interesting findings, it is difficult to distinguish which observations are novel versus already known in the literature regarding maize HM stress responses. The rationale behind focusing on specific loci is often lacking. For example, a statistically significant region identified via LOD score on chromosome 5 contains over 50

genes, yet the authors focus on three known HM-related genes without discussing others in the region. It is unclear why ZmHMA1 was selected for mutagenesis over ZmHMA7 or ZmSKUs5.

(2) The idea that HM stress impacted gene function and influenced human selection during domestication is of interest. However, the data presented do not convincingly link environmental factors with human-driven selection or the paleoenvironmental context of the transition. While lower nucleotide diversity values in maize could suggest selective pressure, it is not sufficient to infer human selection and could be due to other evolutionary processes. It is also unclear whether the statistical analysis was robust enough to rule out bias from a narrow locus selection. Furthermore, the addition of paleoclimate records (Paleoenvironmental Data Sources as a starting point) or conducting ecological niche modeling or crop growth models incorporating climate and soil scenarios would strengthen the arguments.

(3) Despite the interest in examining HM stress in maize and the presence of a pleiotropic phenotype, the assessment of the impact of gene expression is limited. The authors rely on qPCR for two ZmHMA genes and the locus *tb1*, known to be associated with maize architecture. A transcriptomic analysis would be necessary to 1- strengthen the proposed connection and 2- identify other genes with linked QTLs, such as those in the short arm of chromosome 5.

<https://doi.org/10.7554/eLife.105858.1.sa2>

Reviewer #2 (Public review):

Summary:

This work explores the phenotypic developmental traits associated with Cu and Cd responses in *teosinte parviglumis*, a species evolutionarily related to extant maize crops. Cu and Cd could serve as a proxy for heavy metals present in the soils. The manuscript explores potential genetic loci associated with heavy metal responses and domestication identified in previous studies. This includes heavy metal transporters, which are unregulated during stress. To study that, the authors compare the plant architecture of maize defective in ZmHMA1 and speculate on its association with domestication.

Strengths:

Very few studies covered the responses of teosintes to heavy metal stress. The physiological function of ZmHMA1 in maize also gives some novelty in this study. The idea and speculation section is interesting and well-implemented.

Weaknesses:

The authors explored Cu/Cd stress but not a more comprehensive panel of heavy metals, making the implications of this study quite narrow. Some techniques used, such as end-point RT-PCR and qPCR, are substandard for the field. The phenotypic changes explored are not clearly connected with the potential genetic mechanisms associated with them, with the exception of nodal roots. If teosintes in response to heavy metal have phenotypic similarity with modern landraces of maize, then heavy metal stress might have been a confounding factor in the selection of maize and not a potential driving factor. Similar to the positive selection of ZmHMA1 and its phenotypic traits. In that sense, there is no clear hypothesis of what the authors are looking for in this study, and it is hard to make conclusions based on the provided results to understand its importance. The authors do not provide any clear data on the potential influence of heavy metals in the field during the domestication of maize. The potential role of *Tb-1* is not very clear either.

Author response:

Reviewer 1:

The selection of heavy metal stress as the condition to investigate is not speculative. The elucidation of the genome from the *Palomero toluqueño* maize landrace revealed heavy metal effects during domestication (Vielle-Calzada et al., 2009). Differences concordant with its ancient origin identified chromosomal regions of low nucleotide variability that contained the three domestication loci included in this study; all three are involved in heavy-metal detoxification. Results presented in Vielle-Calzada et al 2009 indicated that environmental changes related to heavy metal stress were important selective forces acting on maize domestication. Our study expands those results by starting to elucidate the function of these heavy metal response genes and their role in the evolutionary transition from teosinte *parviglumis* to maize.

Although the paper presents some interesting findings, it is difficult to distinguish which observations are novel versus already known in the literature regarding maize HM stress responses. The rationale behind focusing on specific loci is often lacking. For example, a statistically significant region identified via LOD score on chromosome 5 contains over 50 genes, yet the authors focus on three known HM-related genes without discussing others in the region. It is unclear why ZmHMA1 was selected for mutagenesis over ZmHMA7 or ZmSKUs5.

We appreciate the value of this comment. We will modify the manuscript to clearly show which phenotypic observations are novel and which were previously reported for maize grown under HM stress. The rationale for focusing on three specific loci is related to results from Vielle-Calzada et al. 2009 (see comment above). Although we demonstrated that these three loci show unusual reduction in genetic variability when compared to the rest of chromosome 5 – including a separate class of genes previously identified as being affected by domestication (Hufford et al., 2012) -, we will expand the genetic and expression analysis to all genes included in a region precisely defined via LOD scores of five QTL 1.5-LOD support intervals that overlap with *ZmHMA1*. Within this region of 1.5 to 2 Mb, we will compare nucleotide variability and gene expression in response to HMs. Contrary to major domestication loci showing a single highly pleiotropic gene responsible for important domestication traits, in this chr.5 genomic region phenotypic effects are due to multiple linked QTLs (Lemmon and Doebley, 2014). The mutagenic analysis of *ZmHMA7* and *ZmSKUs5* will be included in a different publication; we can anticipate that the results reinforce the conclusions of this study.

The idea that HM stress impacted gene function and influenced human selection during domestication is of interest. However, the data presented do not convincingly link environmental factors with human-driven selection or the paleoenvironmental context of the transition. While lower nucleotide diversity values in maize could suggest selective pressure, it is not sufficient to infer human selection and could be due to other evolutionary processes. It is also unclear whether the statistical analysis was robust enough to rule out bias from a narrow locus selection. Furthermore, the addition of paleoclimate records (Paleoenvironmental Data Sources as a starting point) or conducting ecological niche modeling or crop growth models incorporating climate and soil scenarios would strengthen the arguments.

We agree that lower nucleotide diversity values in maize are not sufficient to infer human selection and could be due to other evolutionary processes. As a matter of fact, since these same HM response loci also show unusually low nucleotide variability in teosinte *parviglumis* (Fig 2), we cannot discard the possibility that natural selection forces related to environmental changes could have affected native teosinte *parviglumis* populations in the early Holocene, before maize emergence. This possibility supports a speculative model suggesting that phenotypic changes caused by HM stress could have preceded human selection and its consequences, contributing to initial subspeciation; the model is proposed in the “Ideas and Speculation” section of the manuscript. Fortunately, as suggested by the reviewer, a large body of paleoclimatic records and paleoenvironmental data is available for the Trans-Mexican Volcanic Belt in the Holocene, including geographic regions where the emergence of maize presumably occurred. We will include an extensive analysis of available paleoenvironmental data and discuss it at the light of our current results regarding the effects of HM stress. We are also expanding the physical range of our statistical analysis to cover at least 60 Kb per locus - including neighboring genes for all three loci - to determine if our results could be due to narrow locus selection.

Despite the interest in examining HM stress in maize and the presence of a pleiotropic phenotype, the assessment of the impact of gene expression is limited. The authors rely on qPCR for two ZmHMA genes and the locus tb1, known to be associated with maize architecture. A transcriptomic analysis would be necessary to 1- strengthen the proposed connection and 2- identify other genes with linked QTLs, such as those in the short arm of chromosome 5.

Although real-time qPCR is an accurate and reliable approach to assess the expression of specific genes such as *ZMHMA1* and *Tb1*, we will explore the possibility of complementing our analysis with available RNA-seq results that are pertinent for this study (see for example Li et al., 2022 and Zhang et al., 2024) and further explore causative effects between HM stress, *Tb1* and *ZmHMA1* expression. As also pointed by Reviewer#1, TEs are known to influence gene expression under abiotic stress and RNA-Seq analysis would allow to determine if TE activity could lead to similar outcomes.

Reviewer #2:

The authors explored Cu/Cd stress but not a more comprehensive panel of heavy metals, making the implications of this study quite narrow. Some techniques used, such as end-point RT-PCR and qPCR, are substandard for the field. The phenotypic changes explored are not clearly connected with the potential genetic mechanisms associated with them, with the exception of nodal roots. If teosintes in response to heavy metal have phenotypic similarity with modern landraces of maize, then heavy metal stress might have been a confounding factor in the selection of maize and not a potential driving factor. Similar to the positive selection of ZmHMA1 and its phenotypic traits. In that sense, there is no clear hypothesis of what the authors are looking for in this study, and it is hard to make conclusions based on the provided results to understand its importance. The authors do not provide any clear data on the potential influence of heavy metals in the field during the domestication of maize. The potential role of Tb-1 is not very clear either.

Thank you for these comments. We will clearly emphasize our hypothesis that HM stress was an important factor driving the emergence of maize from teosinte *parviglumis* through action of HM response genes. A comprehensive panel of heavy metals would not be more accurate in terms of simulating the composition of volcanic soils evolving across 9,000 years in the region where maize presumably emerged. Copper (Cu) and cadmium (Cd) correspond each to a different affinity group for proteins of the ZmHMA family. ZmHMA1 has preferential affinity for Cu and Ag (silver), whereas ZmHMA7 has preferential affinity to Cd, Zn (zinc), Co

(cobalt), and Pb (lead). Since these P1b-ATPase transporters mediate the movement of divalent cations, their function remains consistent regardless of the specific metal tested, provided it belongs to the respective affinity group. By applying sublethal concentrations of Cd (16 mg/kg) and Cu (400 mg/kg), we caused a measurable physiological response while allowing plants to complete their life cycle, including the reproductive phase, facilitating a comprehensive analysis of metal stress adaptation.

Although real-time qPCR is an accurate and reliable approach to assess gene expression, we agree that RNA-Seq results would improve the scope of the analysis and better assess the role of *Tb1* in relation to HM response (see comments for Reviewer#1). There are two phenotypic changes clearly connected with the genetic mechanisms involved in the *parviglumis* to maize transition: plant height and the number of seminal roots (not nodal roots). We will emphasize these phenotypic changes in a modified version of the manuscript. There is a possibility for HM stress to represent a confounding factor in the selection of maize and not a driving factor; however, if such is the case, we think it is rather unlikely that the real driving factor could have acted through mechanisms not related to abiotic stress or HM response. To address the possibility that HM stress was a cofounding factor, we will extensively analyze genetic diversity and gene expression in all loci containing genes mapping in close proximity to peak LOD scores of all 1.5-LOD support intervals located in chromosome 5 and showing pleiotropic effects on domestication traits (Lemmon and Doebley, 2014). These will also include those mapping in close proximity to *ZmHMA1*. The potential influence of heavy metals in the field is being investigated through the analysis of paleoenvironmental data (see response to Reviewer#1); we will include our results in a modified version of the manuscript.

We thank both reviewers for their detailed revision the manuscript and their pertinent recommendations to improve its presentation and reading.

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<https://doi.org/10.7554/eLife.105858.1.sa0>